april 16, 2026

quick recap

M1 → MSE

data

train → train MSE

test → test MSE

→ K-fold cross-validation

(1) realistic test MSE

(2) choose our tuning parameters in a data-driven way.

ex: degree $d$

$K$

$\lambda$

# Regularized Regression

recall:

- least squares (linear regression)

n observations

$$\sum_{i=1}^{n} \left( Y_i - \hat{Y}_i \right)^2 \rightarrow \text{RSS}$$

(residual sum of squares)

$\hat{B}_0 + \hat{B}_1 X_i$

- find $\hat{B}$'s that minimize RSS

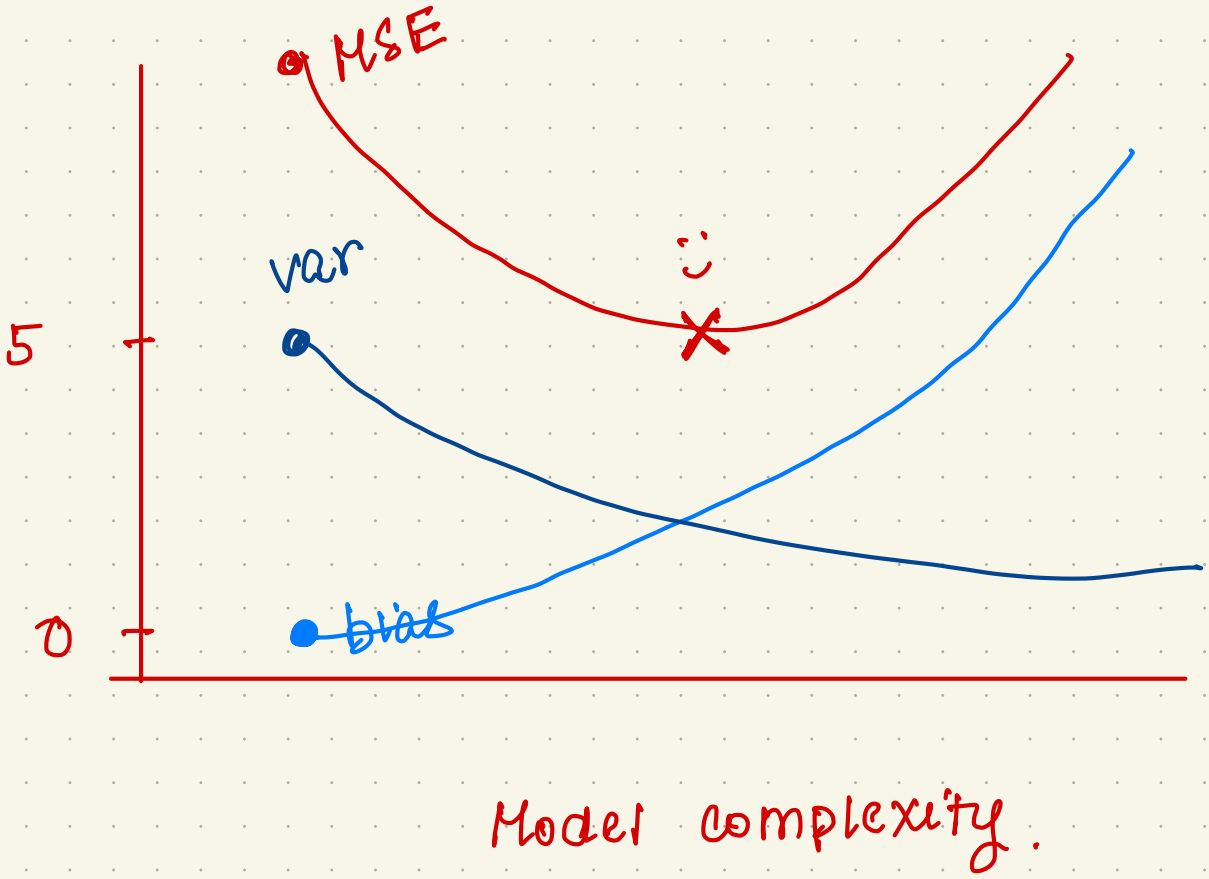
$\hat{B}$ are unbiased.

$$E(\hat{B}) = B$$

$$E(\hat{B}) - B = 0.$$

Bias . variance tradeoff

$$E\left(\text{test MSE}\right) = \text{bias}^2 + \text{var} + (\text{irreducible error})$$




$$RSS + \lambda \text{ penalty}$$

$\uparrow$

bias